



Protein Subcellular Localization Prediction Model Based on Graph Convolutional Network

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Received: 3 February 2022 / Revised: 12 May 2022 / Accepted: 17 May 2022 / Published online: 17 June 2022
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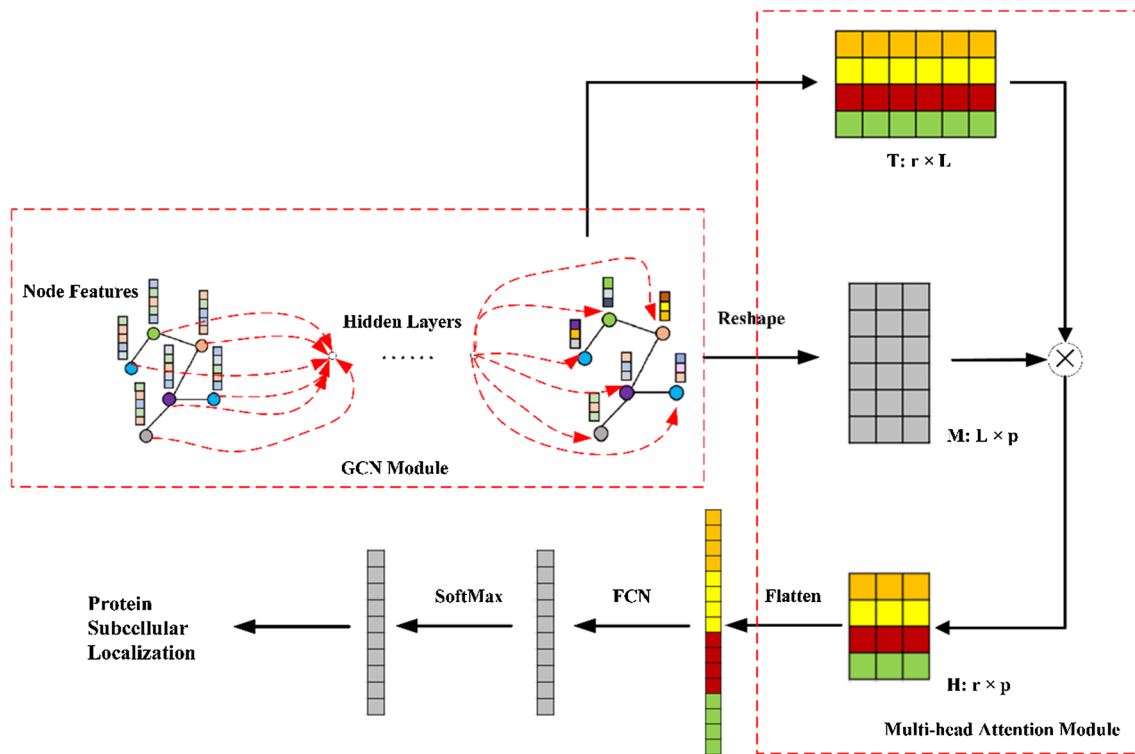
Abstract

Protein subcellular localization prediction is an important research area in bioinformatics, which plays an essential role in understanding protein function and mechanism. Many machine learning and deep learning algorithms have been employed for this task, but most of them do not use structural information of proteins. With the advances in protein structure research in recent years, protein contact map prediction has been dramatically enhanced. In this paper, we present GraphLoc, a deep learning model that predicts the localization of proteins at the subcellular level. The cores of the model are a graph convolutional neural network module and a multi-head attention module. The protein topology graph is constructed based on a contact map predicted from protein sequences, which is used as the input of the GCN module to take full advantage of the structural information of proteins. Multi-head attention module learns the weighted contribution of different amino acids to subcellular localization in different feature representation subspaces. Experiments on the benchmark dataset show that the performance of our model is better than others. The code can be accessed at <https://github.com/GoodGuy398/GraphLoc>.

Graphical abstract

The proposed GraphLoc model consists of three parts. The first part is a graph convolutional network (GCN) module, which utilizes the predicted contact maps to construct protein graph, taking benefit of protein information accordingly. The second part is the multi-head attention module, which learns the weighted contribution of different amino acids in different feature representation subspace, and weighted average the feature map across all amino acid nodes. The last part is a fully connected

layer that maps the flatten graph representation vector to another vector with a category number dimension, followed by a softmax layer to predict the protein subcellular localization.



Keywords Protein subcellular localization · Deep learning · Graph convolutional network · Multi-head attention

1 Introduction

Proteins perform diverse functions in the eukaryotic cells. The functions are limited by the compartments where proteins are located because the compartments provide a physiological environment. If the protein localization is abnormal, its function will be affected and may lead to many diseases, such as Alzheimer's disease, metabolic disorders, and cancers [1]. The mechanisms of protein localization are complex. Locating proteins correctly and studying the localization mechanisms help understand the classification understanding the classification of proteins and designing therapies related to protein localization [2]. Many techniques, such as mass spectrometry [3] and fluorescence-tagging methods [4], have been used to identify the localization of proteins. However, these methods are costly and may generate wrong results. Therefore, only the localization of a limited number of proteins has been identified at the subcellular level. Thus, it is of great significance to predict the subcellular localization of proteins with computational methods [5].

Computational methods for protein localization prediction can be classified into two types. The first type of

methods uses annotations of homologous proteins [6], such as LocTree [7], SherLoc [8], MultiLoc2 [9], and YLoc [10]. These methods perform well if reliable annotations of homologous proteins are available in databases, while the performance will decrease significantly if annotated homologous proteins are not available.

The second type of approaches is machine learning, which learn patterns from collected training data. Some traditional machine learning methods rely on manually extracted features. For example, Wolf psort [11] used features based on amino acid composition, sorting signals and functional motifs to train a KNN classifier for prediction. Based on evolutionary information, Du et al. integrated LDA with SVM to predict the subcellular position of apoptosis proteins [12]. VacPred [13] employed PSSM and SVM for the subcellular localization of vacuole proteins. By detecting the targeting signals near the N-terminal, TPpred3 [14] identified whether the subcellular localization is mitochondrial or chloroplast with an SVM classifier. Similarly, N-terminal sequence information and neural networks were also used in Predotar [15] and TargetP [16]. The former was to identify three subcellular localizations of plastids, endoplasmic reticulum and mitochondria, while the latter focused on the

subcellular localization of chloroplasts, secretory pathways, and mitochondria.

With the rapid development of deep learning, it is also applied to protein localization prediction instead of traditional machine learning methods. For example, DeepLoc [17] employed a recurrent neural network and an attention mechanism to predict protein localization. Long et al. combined a deep convolutional neural network-based feature extractor and a multi-head self-attention encoder to aggregate multiple feature vectors to predict the subcellular locations [18]. MULocDeep [19] predicted protein localization at the organelle and subcellular levels with self-attention mechanism and LSTM. Cong et al. presented a model based on DCNN with a self-attention mechanism for multiple protein subcellular locations prediction [20]. Although these methods have produced good predictions in protein localization, they do not utilize the structural information of proteins.

In the last few years, various deep learning approaches have employed graph neural networks (GNN) to use protein structures and have made successes [21]. These methods require the structures of proteins, which are often expensive to obtain, so in most cases, they are not applicable for predicting protein localization. According to the Critical Assessment of Protein Structure Prediction (CASP) [22], protein structure prediction has significantly been developed in recent years. For instance, SPOT-Contact [23], a sequence-based approach, has been proposed to predict protein contact maps and performs well on the independent test set. The high accuracy of contact map prediction makes it possible to take advantage of protein structure for subcellular localization identification.

This paper proposes a protein subcellular localization prediction model called GraphLoc. This model employs the predicted contact maps to utilize protein structural information, which is processed by the graph convolution network module. Multi-head attention mechanism is also used in the model, learning the weighted contribution of different amino acids to subcellular localization in different feature representation subspaces. Our model achieves good accuracy in the DeepLoc dataset [17], outperforming some current state-of-the-art algorithms.

2 Methods

2.1 Overview

Assuming that a protein sequence consists of L amino acids, it can be represented as a topological map $G(N, E)$, where N is the set of nodes (amino acids), and E is the set of edges between different nodes. A pair of nodes is

connected if it has the top highest contact probabilities in the predicted contact map. The task is to learn the mapping function:

$$F : (G, X) \rightarrow P \quad (1)$$

where G is the above mentioned topological map of the protein, $X \in R^{L \times f}$ denotes the feature matrix of the nodes, and $P \in R^c$ is the subcellular localization probabilities. f is the feature dimension of single amino acid, and c is the number of protein subcellular localization classes, being taken as 10 in this paper. In this work, the mapping F is a graph convolutional neural network model that aggregates information on the nodes and edges of the protein graph.

The proposed GraphLoc model is composed of three components. As illustrated in Fig. 1, the first component is a graph convolutional network (GCN) [24] module, which aggregates feature information from neighbouring nodes during iteration. The second component is the multi-head attention module, which learns the weighted contribution of different amino acids in different feature representation subspaces and performs a weighted average of all amino acid nodes. The last part is a fully connected layer that maps the flattened graph representation vector to another vector with a category number dimension, followed by a SoftMax layer to predict the protein subcellular localization.

2.2 Protein Node Feature Representation

In this paper, three types of features are utilized for training the proposed model. The three types of node features and their dimensions are listed in Table 1. The input X is the size of the sequence length (L) $\times$ the amino acid vocabulary (45).

Blosum62 Blosum62 is a 20×20 substitution matrix widely used for residue encoding [25]. The matrix evaluates all exchanges of an amino acid with another amino acid to measure similarity. It provides a 20-dimensional feature representation for each of the 20 amino acids.

PSSM The PSSM profile is usually generated by comparing multiple sequences against a large database. In our experiments, the PSSM profile is generated by performing PSIblast [26] on the Swissprot database [26] with e-value of 0.001 after three iterations.

Physical-chem Properties Venkatarajan and Braun proposed a comprehensive Physical-chem property set with 237 elements for different types of amino acids [28]. We take the first five eigenvectors of the feature set to improve computational efficiency while maintaining the whole Physical-chem feature set information according to the principle component analysis method.

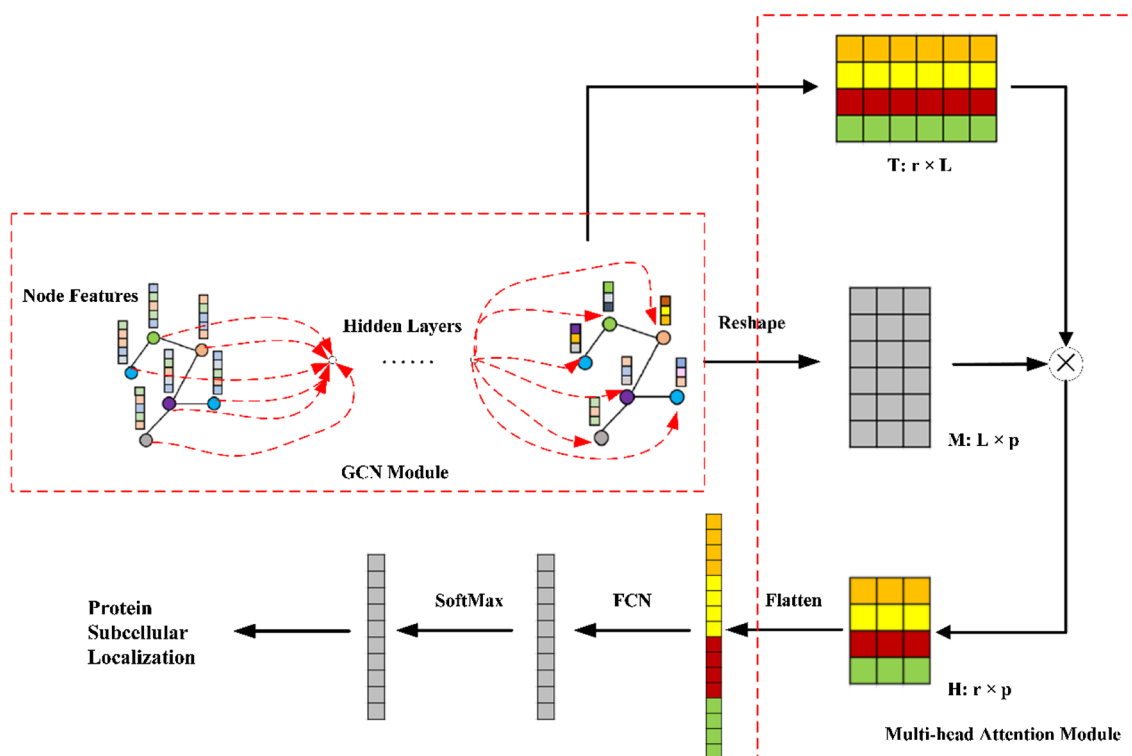


Fig. 1 The framework of GraphLoc

Table 1 Three types of node features and their dimensions

Node features	Dimensions
BLOSUM62	20
Position-specific scoring matrix	20
Physical–chemical properties	5

2.3 Protein Contact Maps

SPOT-Contact [23] is a protein contact map prediction model based on deep learning. It stacks the residual convolution networks with two-dimensional residual bidirectional recurrent LSTM networks, and uses both one-dimensional sequence-based and two-dimensional evolutionary coupling-based information as input features. It has a good performance in CASP and requires less hardware compared with other models. Therefore, our method uses SPOT-Contact to construct the protein graph. For a protein with length L , SPOT-Contact outputs an $L \times L$ matrix, and the values of its elements are the predicted contact probabilities between the corresponding residue pairs. Those pairs with the highest predicted probabilities are actual contacts in the protein graph G . Since the number of actual contacts in proteins

is roughly proportional to the length, we choose a certain percentage of the highest probabilities as the actual contacts. In this paper, it is tested that the top $\alpha \times L$ highest predicted contacts are taken as real contacts. Following the suggestion of Ref [29], α is set from 1 to 7 for comparison.

2.4 GCN Module

For a protein sequence with L amino acids, we denote $X \in R^{L \times f}$ as the feature matrix on protein graph G , which means that each amino acid node has an f -dimensional feature vector. Denote $A \in R^{L \times L}$ as the contact matrix, where the elements of A mean whether the corresponding residue pairs are connected. The GCN module consists of multiple GCN layers, each of which could be described as the following formula:

$$X^{l+1} = \sigma(A'X^lW^l) \quad (2)$$

where $A' = A + I$ is the sum of the adjacency matrix A and the identity matrix I , $X^l \in R^{L \times f}$ is the hidden feature in the l th layer with initial state $X^0 = X$. $W^l \in R^{f \times f'}$ is a trainable weight matrix for mapping feature matrix to a feature space

with dimension f' , σ indicates a non-linear activation function, which is taken as *ReLU* function in this paper. Each GCN layer is followed by a normalization layer that maps the output to $[0, 1]$, which can accelerate the convergence of the GCN layers. The output of the final GCN layer is a matrix $M \in R^{L \times p}$.

2.5 Multi-head Attention Module

The dimension of GCN module output matrix M is associated with the protein length L , which varies with different proteins. It is necessary to eliminate size variance in sequence length to acquire a representation with a fixed size [30]. Therefore, multi-head attention mechanism [31] is used to compute the contribution of each amino acid. Attentions with r heads could be expressed as $T \in R^{r \times L}$:

$$T = \text{SoftMax}(W_2 \tanh(W_1 M^T)) \quad (3)$$

where $W_1 \in R^{q \times p}$ and $W_2 \in R^{r \times q}$ are two learned attention matrices. The *SoftMax* function is defined as

$$\text{SoftMax}(x_i) = \frac{\exp(x_i)}{\sum_j \exp(x_j)} \quad (4)$$

which normalizes each dimension feature according to the computed attention coefficients on different nodes. The r groups of attention coefficients evaluate the contributions of each amino acid to the subcellular localization from different perspectives. Then we multiply T and M and flatten the result into a one-dimensional vector of fixed length $H \in R^{(r \times p)}$:

$$H = \text{Flatten}(TM) \quad (5)$$

2.6 Fully Connected Layer

The output of the multi-head attention module is input to the full connected layer to predict the probability of protein subcellular localization. Through a fully connected layer with weight matrix $W_3 \in R^{c \times (r \times p)}$, the matrix is transformed into a vector with a c dimension, where c is the number of protein subcellular localization types, taken as 10 in this paper. Then the *SoftMax* function is utilized to map the values to the interval $[0, 1]$, meaning the probabilities of different protein subcellular localizations, namely that

$$P = \text{SoftMax}(W_3 H^T) \quad (6)$$

The class with the highest probability is the predicted result.

3 Experiments

3.1 Dataset

3.1.1 DeepLoc dataset

We train and evaluate our model on the DeepLoc dataset [17], taken from the UniProt database, version 201,604. The following criteria were applied when the data were collected: eukaryotic that is encoded in the nucleus, not fragments, experimentally annotated and longer than 40. The similar localizations or subclasses of the same localization were merged into ten main localizations. Finally, 13,858 proteins were collected after all steps. The DeepLoc dataset is a multi-class but not multi-label dataset, meaning that proteins with multiple subcellular localization are deleted. The training and test data are separated with less than 30% identity. Due to the length limitation of the contact map prediction model, we remove the proteins over a thousand amino acids in the DeepLoc dataset. The final training data has 10,038 proteins, while the test data has 2446.

3.2 Cross-Validation and Evaluation indicators

An eight-fold cross-validation method is performed on the DeepLoc training dataset. The proteins in training data are divided into eight folds with similar localization distributions. The proteins in any fold have less than 40% sequence identity with the proteins in the other folds. A model is trained on seven folds and evaluated on the left fold each time. The procedure is repeated eight times, and the results of the eight models are averaged as the validation results and are used to optimize hyperparameters. After optimizing the hyperparameters, a model is trained on all training data and tested on the test data independently. The performance of the model is evaluated by accuracy, precision, recall, MCC, and AUC, the first four of which are defined as:

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

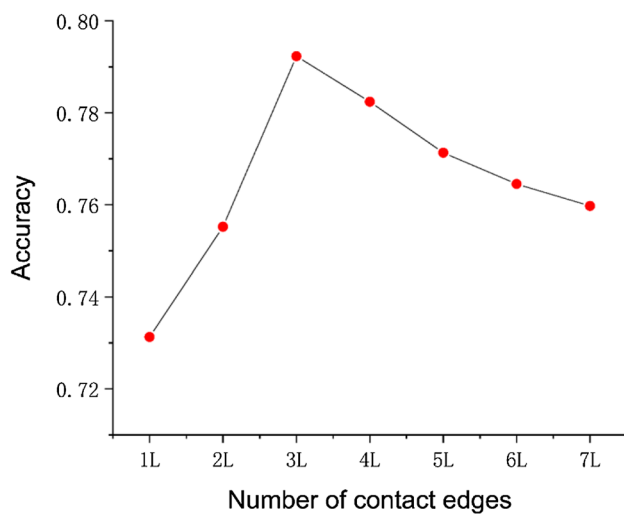


Fig. 2 The relationship between accuracy and the number of contact edges

$$\text{precision} = \frac{TP}{TP + FP} \quad (8)$$

$$\text{recall} = \frac{TP}{TP + FN} \quad (9)$$

$$\text{MCC} = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (10)$$

where TP, TN, FP, FN are True Positives, True Negatives, False Positives, and False Negatives, respectively. The AUC (Area Under Curve) is defined as the area enclosed by the axis below the ROC curve, which could be approximated by

$$\text{AUC} = \frac{\sum_{a \in P} \text{rank}_i - \frac{|P| * (|P| + 1)}{2}}{|P| * |N|} \quad (11)$$

where P denotes the set of positive samples, $|P|$ indicates the number of positive samples, $|N|$ indicates the number of negative samples, and rank_i denotes the ranking of the i th sample according to the prediction score of the model, respectively.

3.3 Hyperparameter Tuning

Our model includes several hyperparameters. We tuned the following hyperparameters:

- (1) *The number of the contact* The number of connected map edges is proportional to the number of residuals, so we test the accuracy in the case of selecting $\alpha \times L$ edges. α is set from 1 to 7 for comparison. This is tested by assigning all selected edges to 1 and other

Table 2 Comparisons of different methods on the DeepLoc test set

Model	Accuracy
LocTree2	0.6120
MultiLoc2	0.5592
SherLoc2	0.5815
YLoc	0.6122
CELLO	0.5521
iLoc-Euk	0.6820
WoLF PSORT	0.5671
DeepLoc	0.7797
MULocDeep variant	0.7790
GraphLoc	0.7923

Note: The highest accuracy value is bolded.

residual pairs to 0. As shown in Fig. 2, the accuracy increases as α increases from 1 to 3, indicating that the predicted contacts contribute to the prediction. After that, the accuracy starts to decline, possibly because of an increase in inaccurate contact prediction.

- (2) *The number of GCN layers* On one hand, many GCN layers imply aggregating the features of higher-order neighbourhoods of nodes. On the other hand, too many layers have the problem of over-smoothing and node features tend to homogenize, which leads to a decrease in prediction accuracy. Hence, it is vital to balance between the complexity of the algorithm and the number of GCN layers. After testing, we find that two layers are optimal.
- (3) *GCN hidden layer size* It determines the channel size in all GCN layers. This hyperparameter impacts the output matrix of the GCN module and thus influences the final prediction. By grid searching, it is found that the optimal parameter for the last layer is 64 and 256 is optimal for the others.
- (4) *The number of attention heads* The attention provides weighting factors to concentrate on the critical residues for localization prediction. Different attention heads evaluate the contribution of each amino acid to localization from different views. It is found that 64 attention heads provide the best performance through experimental validation.

3.4 Performances and Analysis

To compare the effectiveness of our models, we train and test the following protein localization models on the DeepLoc dataset, including LocTree2, MultiLoc2, SherLoc2, YLoc, CELLO, iLoc-Euk, WoLF PSORT, DeepLoc, MULocDeep, and our model GraphLoc, the last three of which are deep learning-based models. MULocDeep is a multi-label model, it could not be applied to the DeepLoc dataset directly.

Table 3 Comparison results on the DeepLoc dataset

Subcellular	Model	MCC	Precision	Recall	Accuracy	AUC
Nucleus	DeepLoc	0.78	0.85	0.84	0.91	0.95
	MULocDeep variant	0.73	0.75	0.88	0.89	0.96
	GraphLoc	0.73	0.75	0.87	0.88	0.95
Cytoplasm	DeepLoc	0.61	0.66	0.71	0.88	0.84
	MULocDeep variant	0.54	0.66	0.56	0.88	0.92
	GraphLoc	0.67	0.80	0.65	0.91	0.93
Extracellular	DeepLoc	0.91	0.91	0.93	0.98	0.98
	MULocDeep variant	0.90	0.90	0.93	0.97	0.98
	GraphLoc	0.87	0.91	0.87	0.97	0.99
Mitochondrion	DeepLoc	0.81	0.85	0.82	0.96	0.98
	MULocDeep variant	0.81	0.84	0.83	0.96	0.98
	GraphLoc	0.82	0.82	0.85	0.96	0.98
Cell membrane	DeepLoc	0.73	0.77	0.74	0.95	0.97
	MULocDeep variant	0.68	0.70	0.73	0.95	0.97
	GraphLoc	0.80	0.84	0.80	0.97	0.97
ER	DeepLoc	0.65	0.66	0.69	0.96	0.93
	MULocDeep variant	0.61	0.63	0.64	0.95	0.93
	GraphLoc	0.68	0.67	0.73	0.96	0.96
Plastid	DeepLoc	0.88	0.86	0.92	0.99	0.99
	MULocDeep variant	0.90	0.90	0.90	0.99	0.99
	GraphLoc	0.90	0.87	0.95	0.99	0.99
Golgi apparatus	DeepLoc	0.41	0.49	0.37	0.97	0.84
	MULocDeep variant	0.49	0.59	0.43	0.98	0.88
	GraphLoc	0.51	0.65	0.41	0.98	0.90
Lysosome	DeepLoc	0.19	0.24	0.19	0.97	0.78
	MULocDeep variant	0.07	0.14	0.05	0.97	0.83
	GraphLoc	0.64	0.74	0.56	0.99	0.95
Peroxisome	DeepLoc	0.32	0.40	0.27	0.99	0.91
	MULocDeep variant	0.38	0.83	0.18	0.99	0.95
	GraphLoc	0.65	0.79	0.54	0.99	0.92
Average	DeepLoc	0.629	0.669	0.648	0.956	0.917
	MULocDeep variant	0.611	0.694	0.613	0.953	0.939
	GraphLoc	0.727	0.784	0.723	0.96	0.954

Note: For each subcellular, the highest value of each performance indicator is bolded.

Therefore, the MULocDeep variant model is used instead, which is provided in Ref. [19]. As DeepLoc is a multi-class dataset, accuracy is used as an evaluation criterion for comparison, which is shown in Table 2. It is easy to find that the last three deep learning-based models significantly outperform the other seven methods. For example, GraphLoc improves by 9.7% over the best of the seven methods, iLoc-Euk. For the three deep learning-based models, GraphLoc performs best, which is 1.2% better than DeepLoc and 1.3% better than the MULocDeep variant, respectively.

For a more accurate inspection of the performance of GraphLoc, it is compared with the second-ranked and the third-ranked methods, namely that DeepLoc and MULocDeep variant on the five metrics of MCC, precision, recall, accuracy and AUC for ten individual localization classes.

Table 3 shows the comparison results, where the bold items are the best results among the three comparison methods. From the table, it could be found that DeepLoc performs best on two subcellular localization classes on the MCC metric, while MULocDeep variant performs best on one subcellular localization class and GraphLoc performs best on eight classes on the MCC metric. Figure 3 shows the number of subcellular classifications that perform best on the five metrics for each method. For precision, recall, accuracy and AUC, the numbers of subcellular localization classes perform best for DeepLoc, MULocDeep variant and GraphLoc are 3, 2, 6; 2, 3, 6; 6, 4, 8; and 3, 5, 8, respectively. Compared with DeepLoc, GraphLoc improves by an average of 9.8%, 11.5%, 7.5%, 0.4% 3.7%, on MCC, precision, recall, accuracy and AUC, respectively. Compared with MULocDeep

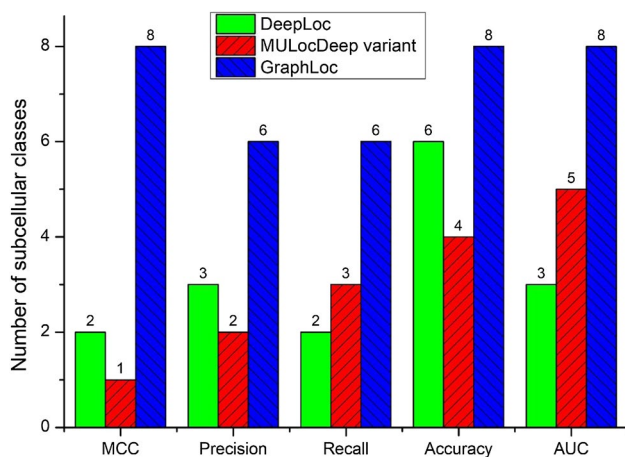


Fig. 3 The number of subcellular classes that perform best on the five metrics

variant, the average improvements of GraphLoc on MCC, precision, recall, accuracy and AUC are 11.6%, 9.0%, 11.0%, 0.7% and 1.5%, respectively. Numerical results show that GraphLoc is clearly superior to both competitors.

In addition, to verify the impact of the contact map accuracy on the model, a comparative experiment is executed using the CCMPred predicted contact map to construct protein graph. The comparison results are shown in Fig. 4. In the figure, light blue, red and dark blue columns are the long-range accuracies of CCMPred and SPOT-Contact in CASP12 on top L/5, L/2 and L levels, which are cited from Ref. [32]. It could be found that the results of CCMPred are significantly lower than those of SPOT-Contact, which decreased by 28.94%, 24.92% and 23.75% at the top L/5, L/2 and L levels, respectively. Consequently, when the protein map is constructed by CCMPred instead of SPOT-Contact, the accuracy of GraphLoc is reduced by 12.92%. It indicates that the accuracy of the contact map affects the performance of the GraphLoc, on the other hand, GraphLoc can narrow the gap between the performance of the predicted contact graphs.

To verify the validity of the three types of features, we perform the localization prediction in the absence of each one of the feature types, respectively, and the experiment results are shown in Table 4. After removing BLOSUM62,

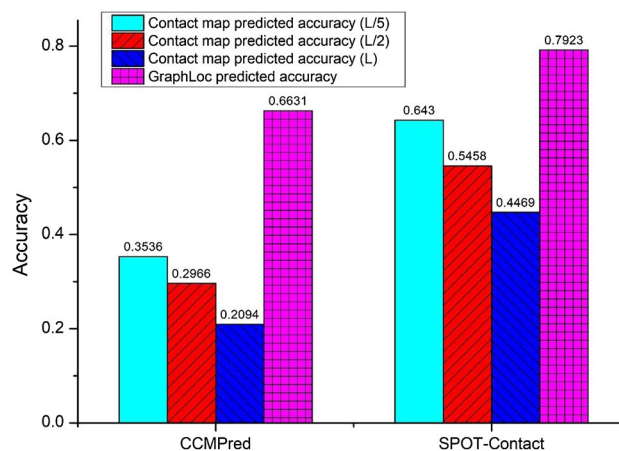


Fig. 4 The comparison results between CCMPred and SPOT-Contact (contact map prediction accuracy (L/5, L/2, L) and GraphLoc protein subcellular localization prediction accuracy using the CCMPred and SPOT-Contact predicted contact map)

Table 4 Comparison of performances for different features

Features	Accuracy
PSSM + physical–chemical properties	0.6909
BLOSUM62 + physical–chemical properties	0.5343
BLOSUM62 + PSSM	0.7076
BLOSUM62 + PSSM + physical–chemical properties	0.7923

Table 5 Comparison of performances for different model architectures

Model	Accuracy
GraphLoc(without GCN)	0.5273
GraphLoc	0.7923

the accuracy decreases by 10.14%. After removing PSSM, the accuracy decreases by 25.8%. After removing physical–chemical properties, the accuracy decreases by 8.47%. The results show that using all the three types of features outperforms the other three cases by removing one type of features, indicating that all three types of features are helpful.

To verify the effectiveness of the GCN module in our model, ablation experiments are executed by removing the GCN module. From Table 5, it can be seen that without the GCN module, the accuracy decreases by 26.5%. Therefore, it could be concluded that the GCN module is necessary.

4 Conclusion

This paper presents the GraphLoc model, a sequence-based subcellular localization prediction algorithm. The innovation of the model is the combination of graph convolutional network and attention mechanism, which effectively utilizes protein structure information through the predicted protein contact graph. With predicted contact maps, this model fully utilizes the protein structural information. Experiments show that the proposed algorithm outperforms some current state-of-the-art algorithms on the benchmark dataset. In the future, the method could be helpful in the prediction of protein function and drug design.

Acknowledgements This research was funded by the National Natural Science Foundation of China (61972174), Key research and development project of Jilin Provincial Science and Technology Department (20210201080GX), Jilin Province Development and Reform Commission (2021C044-1), Guangdong science and technology planning (2020A0505100018), Guangdong universities' innovation team (2021KCXTDO15), and Guangdong key disciplines (2021ZDJS138) projects.

Declarations

Conflict of Interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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